Assignment -1

1. Raspberry Pi (Gen 2)

The Raspberry Pi is basically a small board that acts like a computer but costs way less than a normal one. It is popular for things like coding projects and robotics or even IoT setups I think. People connect sensors and devices to it using the GPIO pins along with USB ports and HDMI output.

The system runs on Raspberry Pi OS which comes from Linux. It has an ARM processor too and that keeps everything affordable for beginners. A lot of students buy one to learn programming without spending much. It shows up in home automation and education stuff quite often.

Maybe the GPIO part is what makes it stand out for electronics work. Some people see it one way and others do not but it seems useful anyway. That part gets a bit messy if you try to list every use.

2. Attending the AI Summit, the following are the top 10 takeaways from the day.

AI is being utilized in writing, designing, research or learning fields. But experts stressed the responsible, safe and ethical use of AI. The main takeaway from the summit was the importance of having good data – if you have bad data you get bad results.

The healthcare, banking, educational, and manufacturing sectors are just a handful of industries that are undergoing a transformation with the help of Artificial Intelligence (AI).

A good programmer has a solid understanding of mathematics, and it's essential for anyone aiming to pursue a career in AI to master these programming and math skills. Cloud computing has a significant role to play, and it is the provision of computing power that is needed for training AI models. Apart from that, there is a rise in the number of open-source AI tools and making AI technology even more accessible. The most important thing to take away from all of this is that AI is rapidly evolving, and that continuous learning is a must.

3. Yotta and NPTEL

Yotta is an Indian company that offers data center and cloud computing services. It provides cloud storage, machine learning infrastructure and GPU compute for enterprise-level organizations that need high-performance computing. The National Programme on Technology Enhanced Learning (NPTEL) is an online learning platform by IIT's and IISc. It offers quality education in the areas of engineering, science, management and other fields via online lectures and certification courses.

4.While selecting a language for an application—what should be taken into

Which language to use for the program is dependent on the needs of the application. Prior to choosing a language, developers should take into account: The nature of the application to be developed.

- Quality of service and performance. The platform that the program is designed for (Web, Android, Windows, Linux, etc.).
- Library space and community resources.
- Security requirements.
- Development cost.
- Programming skills of the development team Examples of this are: AI, Machine Learning, Data Science – Python. Java Web – Web-based applications. Android application development, is done in the following languages: Kotlin. C/C++ – system software, game software, low level programming. JavaScript – Web development.

5. The memory, Word Size, 64-bit Systems, Speed, and Operating System

Random Access Memory (RAM): Memory for temporary storage of data when programs are running. The typical capacities for common RAMs are 8 GB, 16 GB and 32 GB.
Word Size is the number of bits that the processor can manipulate at a time. Common word sizes include 8-bit, 16-bit, 32-bit, and 64-bit. A 64-bit processor, with a 64-bit operating system, will be able to work with more data per cycle and use much more memory than a 32-bit processor, running a 32-bit operating system. The speed of a processor is measured in terms of cycles per second (MHz or GHz – both mean millions, but GHz is billion and MHz is million).
Operating System (OS): controls hardware and software resources of computer systems. Popular operating systems are Windows, Linux and macOS.

6. Why is a GPU Faster than a CPU?

A CPU is made with a few high-performance cores that can process a set of instructions one after the other. Note that, by contrast, GPUs have thousands of smaller cores that can execute many calculations at once. The parallel processing power of GPUs is especially useful for:

- Graphics rendering
- Video processing
- Artificial Intelligence
- Machine Learning
- Scientific computing
- Large-scale mathematical calculations

7. CPU, GPU, and TPU

The Central Processing Unit (CPU). Used to perform general use computing tasks and operate an operating system and common applications. A computer chip used to process graphics and text data.Graphics and text data-processing chip. Designed for graphics processing and parallel computation, making it ideal for AI, gaming, and video rendering. TPU (Tensor Processing Unit) An AI & Machine Learning workloads processor.

8. ACM A.M. Turing Award

Andrew G. Barto and Richard S. Sutton were awarded the 2024 ACM A.M. Turing Award (announced in 2025), for their groundbreaking advances in Reinforcement Learning. Reinforcement Learning allows the computer to learn through the receipt of rewards for correct actions, and penalties for incorrect actions, thus allowing the AI system to learn through experience.

9. Stampeding Herd Problem

A Stampeding Herd Problem is one in which many computers and/or users are simultaneously trying to get something done from a server, which uses a lot of CPU and makes the system perform poorly. Common solutions include:

- Caching
- Delayed retries
- Request queues
- Load balancing
- Rate limiting

10. The creators of Intel and AMD.The inventors of Intel and AMD.

Intel

- Founded in 1968 Founders: Robert Noyce, Gordon Moore AMD
- Advanced Micro Devices is a computer hardware firm.
- Founded in 1969 Founded by: Jerry Sanders and 7 other co-founders.

11. Laptop CPU Architecture:Laptop CPU Architecture:

Most laptops currently on the market have two types of processors: x86-64 – Used by Intel Core processors, and AMD Ryzen processors. ARM64 – For Apple Silicon (M-series) processors. To find the architecture of your processor: Under Windows: Visit Settings →

System → About On Linux: In a terminal window, type in ls cpu. On MacOS: In the Terminal type uname -m.

12. The Imitation Game (2014) Movie Review.

I saw the movie The Imitation Game and it was better than I thought it would be. It is about cracking the German code in world war two with Alan Turing and the others on his team. His determination stands out the most because he did not give up easily on something that big.

Teamwork is another thing that comes up since these kinds of problems need more than one person to figure out. I realized this film could help explain how computers solve actual problems that affect people. It also showed the unfair way Turing was treated after everything he did which does not seem right at all.

Maybe that is the part that stands out the most in the end. The whole thing kind of inspired me but there are some details that get a bit messy when you think about them.